



Selective Auditory Attention Decoding in Naturalistic Conversations Using EEG-Based Speech Envelope Tracking in Multi-Speaker Environments

Gabriel Ivucic¹, Saurav Pahuja², Dashanka Da Silva², Tanja Schultz¹

¹Cognitive Systems Lab, University of Bremen, Germany

²Machine Listening Lab, University of Bremen, Germany

ivucic@uni-bremen.de

Abstract

In daily life, we effortlessly switch attention between different sound sources while filtering out distractions. Prior research has shown that the target speaker can be decoded from EEG in such settings, however, how the switching of attention influences decoding has not been measured. We investigated EEG-based target speaker decoding with exogenous attention switches in a multi-talker environment, where twenty participants alternated listening to two target speakers taking turns while ignoring two background speakers. Speech envelope reconstruction from EEG showed higher accuracy for the attended speaker than distractors, with single-trial classification at 76%. Our findings showed a brief increase in target speaker reconstruction accuracy after attention switches, suggesting heightened alertness when shifting attention to a new speaker. These findings enhance the understanding of dynamic auditory attention and support the development of attention-based hearing applications.

Index Terms: EEG, Auditory Attention, Speech Envelope, Attention Switching

1. Introduction

Navigating speech in noisy environments, such as a crowded room with overlapping conversations, poses a significant challenge for the auditory system, a phenomenon famously referred to as the *cocktail-party problem* [1]. The cognitive ability to focus on a single speaker while suppressing competing voices, known as selective auditory attention [2, 3], is essential for effective communication in everyday life.

Auditory neuroscience seeks to understand how the brain achieves this remarkable ability by studying how acoustic features elicit specific cortical responses. Prior research has investigated neural representations of various speech features, such as spectrograms [4], phonemes [5], and semantic features [6]. Among these, the speech envelope—the slow amplitude fluctuations of speech—has proven particularly effective for studying auditory attention. Research has shown that the reconstructed speech envelope derived from neural data correlates more strongly with the envelope of attended speech than with that of unattended speech, making it a key mechanism for decoding selective attention [7]. Additionally, the speech envelope predominantly consists of low-frequency fluctuations (<10 Hz), which align with cortical oscillations in the delta and theta bands—frequencies known to be crucial for both speech processing and attention [8].

To model the relationship between acoustic features and neural activity, one widely used approach is the temporal response function (TRF) [9], which characterizes how neural responses evolve over time in response to auditory input. This approach

can also be applied in the reverse direction, using what are known as backward models (EEG-to-stimulus reconstruction). These backward models have been shown to outperform forward models (stimulus-to-EEG mapping) in predicting attentional focus [10]. In this context, backward models can be applied to speech envelope reconstruction for auditory sources. As a result, speech envelope reconstruction from EEG has become an effective tool for classifying the attended speaker and comparing attentional dynamics across experimental conditions [11, 12].

However, most research to date has focused on neural responses to speech under static conditions, where participants maintain attention on a fixed speaker at a fixed location. More recent work has explored neural dynamics during attention switches when the attended stimulus changes. One approach has investigated exogenously driven switches, where speaker locations are abruptly swapped to simulate head movements [13], while another has examined endogenous switches, in which participants receive advance cues but must internally initiate the switch themselves [14]. Both found that decoding of stimulus features remains robust despite shifts in attention.

Building on this research, we designed a selective attention experiment that simulates a natural conversation between two speakers embedded in a noisy environment with two additional distractor speakers. Neural responses were recorded using EEG, making our approach comparable to a prior study using invasive EEG [15]. Further, we aimed to maximize the number of occurring attention switches and introduce a greater variety of speakers, using six target speakers and six distractor speakers, as opposed to the typical male-versus-female setup used in previous research. In this study, we aim to verify the robustness of EEG-based speech envelope tracking in the context of following a conversation in a multi-speaker environment. Additionally, we assess the direct impact of exogenous attention switching on target speaker reconstruction accuracy, providing insights into how dynamic shifts in auditory attention affect neural tracking of speech.

2. Materials and Methods

2.1. Experiment for Selective Auditory Attention and Attention Switching

We designed an experiment to simulate a multi-talker environment and investigate how auditory attention switches affect neural processing. EEG was recorded while participants selectively attended to alternating speakers in the presence of continuous background speech.

2.1.1. Setup

Twenty participants listened to one of two target speakers from loudspeakers positioned at -30° (left) and 30° (right) while ignoring two continuous background distractors at -90° and 90° . All loudspeakers were arranged in a semicircle at a distance of 60 cm. EEG data were recorded at 512 Hz using a 64 active-channel system (g.HIAMP, g.tec medical engineering GmbH, Austria). Electrodes were positioned according to the 10/20 system, with two linked-ear reference electrodes and a ground electrode at Fz. Channel impedances were kept below 20 k Ω .

2.1.2. Audio Stimuli

The target speech was drawn from two audiobooks, "20,000 Leagues Under the Sea" [16] and "The Chancellor" [17], narrated by six male speakers (three per book). Each speaker's recordings were divided into segments of approximately 18 seconds each, covering 2 to 3 sentences of continuous speech. Background distractors consisted of six additional male narrators, with whole segments lasting approximately 72 seconds. Audio levels of all speakers were normalized to -23 LUFS, and prolonged silences exceeding 500 ms were truncated.

To ensure balanced exposure, we used three fixed cocktail party configurations, each featuring a unique set of two target and two distractor speakers. Each configuration was presented in 12 trials, totaling 36 trials. The trial order was counterbalanced to prevent adaptation effects, with randomized starting positions of target speakers (left or right). After each trial, participants answered two multiple-choice questions (MCQ) on speech segments selected in a pseudo-random but controlled manner to verify attentional engagement.

2.1.3. Experimental Procedure

Each trial consisted of four $18s \pm 2$ speech segments, with two from each target speaker alternately presented, totaling 72s per trial and resulting in three attention switches. To mimic turn-taking in a natural conversation, target speakers alternated speaking with 300 ms pauses between their speech segments, reflecting typical silence durations in spoken dialogue [18]. However, the content between target speakers was not semantically related (e.g., no question-answering) to prevent multiple-choice questions for certain segments from being answered through information transfer. Throughout a trial, three speakers were always active — one target speaker and two distractors, while the two distraction speakers were constantly talking in the background without the speaker switching. In total, participants completed 36 trials and experienced 108 exogenous attention switches. The entire experimental session lasted around 90 minutes, with approximately 42 minutes of recorded EEG data from the trials.

2.2. Speech Envelope Reconstruction from EEG

We aim to reconstruct the audio envelope, representing the loudness levels of the speech stream, based on its linear relationship with EEG activity. By comparing the reconstruction accuracy for each speaker's envelope, we can decode the target speaker by selecting the envelope with the highest reconstruction accuracy [7].

EEG preprocessing involved the removal of eye blinks using independent component analysis (ICA) and the interpolation of bad channels. Furthermore, data were bandpass filtered between 0.5 and 12 Hz, as this frequency range was found to capture neural modulations associated with the speech envelope

most effectively. [9, 19], and downsampled to 64 Hz. Regarding speech stimuli, for each trial, the attended (target) and unattended (distraction) speech streams were separated into their original clean forms, and their broadband envelopes were extracted using the Hilbert transform. The envelopes were then anti-aliased, downsampled to 64 Hz, and normalized between 0 and 1.

To reconstruct the envelopes from the EEG, we use a linear decoding model that maps the neural responses from all 62 EEG channels over 0–200 ms time lags to the envelope samples that elicited them. We implement this using the receptive field class in the MNE library [20] with multivariate linear regression and ridge regularization. The reconstruction is formulated as:

$$\hat{s}(t) = \sum_{n=1}^{62} \sum_{\tau=0}^{200 \text{ ms}} r_n(t + \tau) w_n(\tau)$$

where $\hat{s}(t)$ is the reconstructed signal envelope at time t , $r_n(t + \tau)$ is the EEG response at channel n and time lag τ , and $w_n(\tau)$ is the corresponding weight. The weights are optimized by solving a regularized least squares problem:

$$\mathcal{L}(w) = \sum_t (s(t) - \hat{s}(t))^2 + \lambda \sum_{n,\tau} w_n(\tau)^2$$

where $s(t)$ is the original envelope. We evaluated the model using leave-one-trial-out cross-validation, using a fixed ridge regularization with $\alpha = 10^2$ [21]. The reconstruction performance of the envelope is measured by calculating the Pearson correlation between the reconstructed and the original envelope. Finally, to decode the attended speaker, we compute the Pearson correlation for each speech stream. Classification accuracy is determined by counting the trials where the attended stream shows a higher correlation than the unattended ones ($r_{attended} > r_{unattended}$) out of all 36 trials, or for respective windows extracted from a trial [10].

3. Results

3.1. Envelope Reconstruction from Neural Responses

We first present the raw envelope reconstruction results from neural responses. Figure 1 illustrates the distribution of Pearson's correlations to reconstruct the target speaker stream (alternating between two speakers) and the constant background distractors across all trials and subjects. For each participant, the target speaker envelope (mean $r = 0.068 \pm 0.045$) was significantly better reconstructed than the distractor speakers ($r = 0.009 \pm 0.03$, $p < 10^{-84}$, Wilcoxon), and also better than a baseline trained with reversed envelope labels (gray box, $p < 10^{-94}$, Wilcoxon). The distractors also showed a slightly higher correlation than the random baseline over all subjects, with a few exceptions ($p < 0.002$, Wilcoxon).

3.2. Target Speaker Decoding

We further evaluate the ability to perform single-trial decoding from the envelopes reconstructed from neural responses. By counting the number of trials ($n=36$) in which the Pearson r for the target speaker exceeded that of the distractor speakers, we obtained an average target speaker detection accuracy of $76\% \pm 12\%$ (chance level: 33%; upper simulated confidence limit: 46.7% [22]), as shown in Figure 2. Notably, participant 10 exhibited poor performance under the significance threshold and was excluded from further analyses. To test the capability

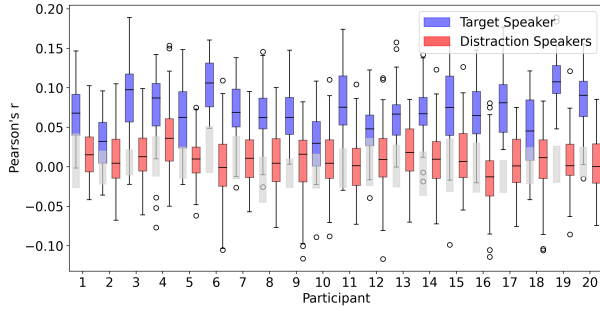


Figure 1: *Speech Envelope Reconstruction results for each participant, based on all 36 trials in a leave-one-out validation. The box plots represent the distribution of pearson correlations for target speakers (blue), distractor speakers (red), and baseline models trained on reversed speech envelopes (gray).*

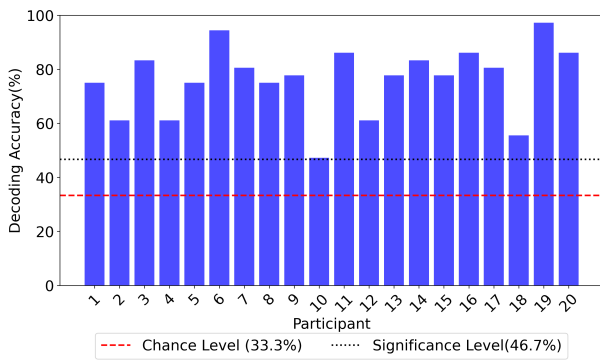


Figure 2: *Single-trial target speaker decoding results based on reconstructed envelopes, across all participants. The significance level corresponds to the upper confidence interval of a binomial distribution ($n=36$).*

of the decoding approach with smaller window sizes than the entire trial, we also inferred the target speaker results for decreasing window sizes, ranging from sliding 20s to 2s windows with a 50% overlap. These results, shown across all participants in Figure 3, demonstrate that for all window sizes, decoding accuracy remains above the significance level, ranging from an average accuracy of 80% down to 44%. Accuracy decreases as the number of decision windows increases, with both the significance threshold and the decision window performance declining as the window size decreases.

To test the capability of the decoding approach with smaller window sizes than the entire trial, we also inferred the target speaker results for decreasing window sizes, ranging from sliding 20s to 2s windows with a 50% overlap. Figure 3 shows that as the window length decreases, decoding performance declines, ranging from an average accuracy of 80% down to 44%. Nevertheless, in all cases, accuracy remains above the significance level, as the significance threshold decreases with the increase in the number of decision windows [22].

3.3. Reconstruction Accuracy around Attention Switching

Finally, we evaluated the direct effect of attention switching on envelope reconstruction performance. First, we extracted ± 10 -second windows around each attention switch, yielding 108 windows per subject. Then, using the previously analyzed 5-second windows, we increased the overlap to 90% (0.5s shift)

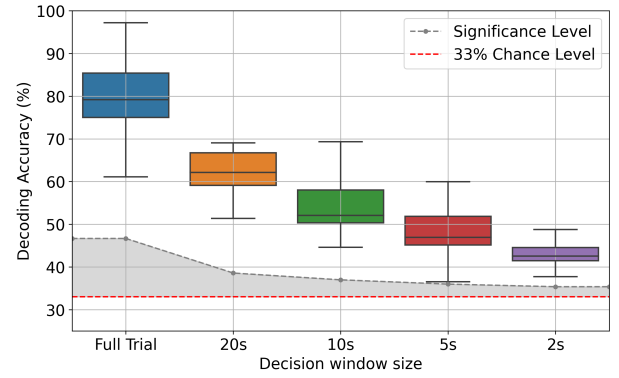


Figure 3: *Target speaker decoding results for all participants across decreasing decision window sizes (50% overlap). The gray dashed line represents the upper confidence interval of a baseline with randomly assigned labels, determined by the number of decision windows.*

to achieve higher temporal resolution. We examined the corresponding correlation values over time, where a window from $t - 5s$ to t represents the correlation value at time point t . Figure 4 shows the averaged Pearson correlation between the reconstructed and original envelopes over all participants, using 0.5s temporal increments. The results indicate weaker envelope tracking before the switch, followed by an approximately 10% increase in reconstruction performance compared to the average correlation, shortly after the switch. Performance returns to baseline levels around 10 seconds post-switch, with the strongest effect observed within ± 6 seconds of the switch. A paired t-test confirmed a significant increase in envelope reconstruction performance before vs. after a switch event (t -statistic = -3.8787 , p -value < 0.0001). Exploratory analyses found a similar trend for the distractor speaker, but this effect was not statistically significant (t -statistic = -1.8254 , p -value = 0.0679) and was therefore omitted. Further, no changes in target speaker decoding accuracy were observed concerning switch timing; therefore, no additional results are reported.

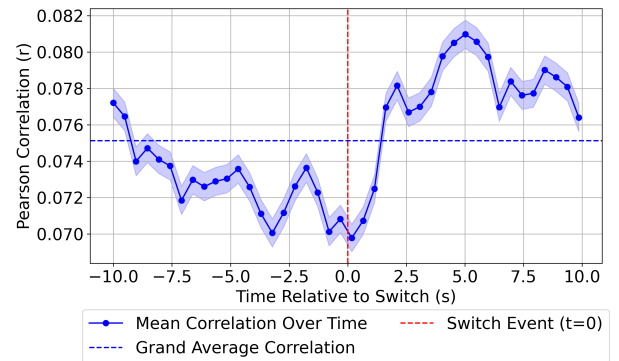


Figure 4: *Envelope reconstruction performance (Pearson's r) ± 10 seconds around switch events, for 5-second windows with a 0.5-second shift, averaged across all participants. The shaded area represents the standard error.*

4. Discussion

For this study, we designed a selective auditory attention experiment that simulates following a conversation to explore the

dynamics of exogenous attention switching and its effects on the neural tracking of speech envelopes from attended speakers. Our results demonstrate that even in the presence of attention shifts, the target speech envelope can still be reliably tracked, with accuracy consistently outperforming that of distractor speakers in single trials. This provides further evidence supporting the utility of envelope reconstruction for decoding auditory attention [7]. Our findings align with recent studies [13, 14, 15] that also observed successful envelope tracking during attention shifts.

We found that decoding remained above chance level even with significantly reduced window sizes, dropping from 80% accuracy to 44% as window sizes were decreased to just two seconds. This performance reduction is consistent with prior evaluations of similar speaker separation angles and window sizes [23, 24]. The drop in accuracy for lower window sizes can likely be attributed to the low signal-to-noise ratio (SNR) of the neural responses to speech signals in EEG data. Notably, our envelope reconstruction results are relatively low compared to intracranial data [25, 26], which offers much higher signal quality. This issue becomes more pronounced in noisy environments where clean speech sources are unavailable [23].

Our evaluations of the effect of attention switching on reconstruction performance revealed a brief increase in decoding accuracy immediately following the switch, followed by a gradual decrease over time, leading up to the next switch. This suggests that neural modulation of the speech envelope is dynamic and responsive to changes in the auditory environment. One possible explanation for this pattern is increased attentional focus following the switch, with listeners paying closer attention to the new target speaker immediately after the attentional shift. This is consistent with previous neuroscience findings, which report a decrease in alpha power in parietal and frontal brain regions shortly after attention shifts, returning to baseline after approximately five seconds [14], typically associated with heightened attentional focus. Interestingly, while we observed a similar pattern for distractor speakers, the effects were not statistically significant. However, these fluctuations in attentional focus did not significantly affect decoding accuracy for the target speaker. Future work could more closely analyze changes in the EEG signal related to attention-switching events.

Our study focused on exogenous attention shifts, such as following a conversation, rather than voluntary, endogenous attention switches, which have been the focus of previous studies [13]. The time lag in our used decoding windows remains a significant challenge for real-time applications, limiting the practicality of such systems in brain-computer interface (BCI) technologies. While inducing and timing endogenous shifts in a controlled environment is challenging, it could offer the advantage of reducing the time lag in decoding, thereby improving real-time performance. This is in line with neuroscientific findings that suggest endogenous attention shifts are linked to top-down preparatory activity [27]. Tracking this preparatory activity could provide a more proactive form of decoding, in contrast to the reactive nature of exogenous shifts.

Looking ahead, future research could focus on directly decoding the moment of attention switching in real-time, enabling neuro-steered hearing aids that adapt to the listener's attention shifts. This could involve leveraging EEG-based features, such as alpha-power modulations, as well as other markers like pupil diameter, which has been shown to correlate with attention switching [14, 28]. Further improvements in decoding accuracy could be achieved by employing more advanced models, such as deep neural networks (DNNs) [19], though these mod-

els require extensive pretraining with large datasets of speakers and samples.

5. Conclusion

This study investigated the dynamics of exogenous attention switching in the context of following a conversation in a noisy environment and its impact on neural tracking of the speech envelope from attended speakers. Our results demonstrate that the target speaker can be reliably decoded even when attention switches between speakers. Additionally, we observed a brief improvement in envelope reconstruction following a switch, likely reflecting a transient increase in attentional focus when attending to a new speaker. While improvements in robust real-time decoding capabilities are needed, these findings lay the groundwork for future research focused on decoding attentional switches, with potential applications in neuro-steered hearing aids for noisy environments.

6. Acknowledgements

This work is supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – GRK2739/1 – Project Nr. 447089431 – Research Training Group: KD²School – Designing Adaptive Systems for Economic Decisions.

7. References

- [1] E. C. Cherry, "Some experiments on the recognition of speech, with one and with two ears," *The Journal of the acoustical society of America*, vol. 25, no. 5, pp. 975–979, 1953.
- [2] A. S. Bregman, *Auditory scene analysis: The perceptual organization of sound*. MIT press, 1994.
- [3] A. W. Bronkhorst, "The cocktail party phenomenon: A review of research on speech intelligibility in multiple-talker conditions," *Acta acustica united with acustica*, vol. 86, no. 1, pp. 117–128, 2000.
- [4] N. Mesgarani and E. F. Chang, "Selective cortical representation of attended speaker in multi-talker speech perception," *Nature*, vol. 485, no. 7397, pp. 233–236, 2012.
- [5] G. M. Di Liberto, J. A. O'sullivan, and E. C. Lalor, "Low-frequency cortical entrainment to speech reflects phoneme-level processing," *Current Biology*, vol. 25, no. 19, pp. 2457–2465, 2015.
- [6] M. P. Broderick, A. J. Anderson, G. M. Di Liberto, M. J. Crosse, and E. C. Lalor, "Electrophysiological correlates of semantic dissimilarity reflect the comprehension of natural, narrative speech," *Current Biology*, vol. 28, no. 5, pp. 803–809, 2018.
- [7] J. A. O'Sullivan, A. J. Power, N. Mesgarani, S. Rajaram, J. J. Foxe, B. G. Shinn-Cunningham, M. Slaney, S. A. Shamma, and E. C. Lalor, "Attentional selection in a cocktail party environment can be decoded from single-trial EEG," vol. 25, no. 7, pp. 1697–1706. [Online]. Available: <https://doi.org/10.1093/cercor/bht355>
- [8] H. Luo and D. Poeppel, "Phase patterns of neuronal responses reliably discriminate speech in human auditory cortex," *Neuron*, vol. 54, no. 6, pp. 1001–1010, 2007.
- [9] M. J. Crosse, G. M. Di Liberto, A. Bednar, and E. C. Lalor, "The multivariate temporal response function (mTRF) toolbox: A MATLAB toolbox for relating neural signals to continuous stimuli," vol. 10, publisher: Frontiers. [Online]. Available: <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2016.00604/full>
- [10] D. D. E. Wong, S. A. Fuglsang, J. Hjortkjær, E. Ceolini, M. Slaney, and A. de Cheveigné, "A comparison of regularization methods in forward and backward models for auditory attention decoding," vol. 12, publisher: Frontiers. [Online]. Available: <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2018.00531/full>

- [11] A. Bednar and E. C. Lalor, "Where is the cocktail party? decoding locations of attended and unattended moving sound sources using eeg," *NeuroImage*, vol. 205, p. 116283, 2020.
- [12] F. Ahmed, A. R. Nidiffer, and E. C. Lalor, "The effect of gaze on EEG measures of multisensory integration in a cocktail party scenario," vol. 17, publisher: Frontiers. [Online]. Available: <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2023.1283206/full>
- [13] E. S. Teoh and E. C. Lalor, "Eeg decoding of the target speaker in a cocktail party scenario: considerations regarding dynamic switching of talker location," *Journal of neural engineering*, vol. 16, no. 3, p. 036017, 2019.
- [14] S. Haro, H. M. Rao, T. F. Quatieri, and C. J. Smalt, "Eeg alpha and pupil diameter reflect endogenous auditory attention switching and listening effort," *European Journal of Neuroscience*, vol. 55, no. 5, pp. 1262–1277, 2022.
- [15] V. Choudhari, C. Han, S. Bickel, A. D. Mehta, C. Schevon, G. M. McKhann, and N. Mesgarani, "Brain-controlled augmented hearing for spatially moving conversations in multi-talker environments," *Advanced Science*, vol. 11, no. 41, p. 2401379, 2024.
- [16] J. Verne, "Der chancellor," LibriVox audio recording, 2010, available at: <https://librivox.org/der-chancellor-by-jules-verne/>.
- [17] —, "20.000 meilen unter dem meer," LibriVox audio recording, 2010, available at: <https://librivox.org/zwanzigtausend-meilen-unterm-meer/>.
- [18] A. S. Meyer, "Timing in conversation," *Journal of Cognition*, vol. 6, no. 1, 2023.
- [19] M. Thornton, D. Mandic, and T. Reichenbach, "Robust decoding of the speech envelope from eeg recordings through deep neural networks," *Journal of neural engineering*, vol. 19, no. 4, p. 046007, 2022.
- [20] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, and M. S. Hämäläinen, "Meg and eeg data analysis with mne-python," *Frontiers in Neuroscience*, vol. 7, no. 267, pp. 1–13, 2013. [Online]. Available: <https://doi.org/10.3389/fnins.2013.00267>
- [21] Z. Fu, X. Wu, and J. Chen, "Congruent audiovisual speech enhances auditory attention decoding with eeg," *Journal of neural engineering*, vol. 16, no. 6, p. 066033, 2019.
- [22] G. Müller-Putz, R. Scherer, C. Brunner, R. Leeb, and G. Pfurtscheller, "Better than random: a closer look on bci results," *International journal of bioelectromagnetism*, vol. 10, no. 1, pp. 52–55, 2008.
- [23] N. Das, A. Bertrand, and T. Francart, "Eeg-based auditory attention detection: boundary conditions for background noise and speaker positions," *Journal of neural engineering*, vol. 15, no. 6, p. 066017, 2018.
- [24] S. Geirnaert, T. Francart, and A. Bertrand, "An interpretable performance metric for auditory attention decoding algorithms in a context of neuro-steered gain control," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 1, pp. 307–317, 2019.
- [25] J. Kubanek, P. Brunner, A. Gunduz, D. Poeppel, and G. Schalk, "The tracking of speech envelope in the human cortex," *PloS one*, vol. 8, no. 1, p. e53398, 2013.
- [26] J. S. Brumberg, D. J. Krusienski, S. Chakrabarti, A. Gunduz, P. Brunner, A. L. Ritaccio, and G. Schalk, "Spatio-temporal progression of cortical activity related to continuous overt and covert speech production in a reading task," *PloS one*, vol. 11, no. 11, p. e0166872, 2016.
- [27] A. K. Lee, S. Rajaram, J. Xia, H. Bharadwaj, E. Larson, M. S. Hämäläinen, and B. G. Shinn-Cunningham, "Auditory selective attention reveals preparatory activity in different cortical regions for selection based on source location and source pitch," *Frontiers in neuroscience*, vol. 6, p. 190, 2013.
- [28] A. Favre-Felix, C. Graversen, R. K. Hietkamp, T. Dau, and T. Lunner, "Improving speech intelligibility by hearing aid eye-gaze steering: Conditions with head fixated in a multitalker environment," *Trends in hearing*, vol. 22, p. 2331216518814388, 2018.